

FINGUARD

Predict. Explain. Simulate. Prevent.

Detect suspicious accounts
Trace money trails. Empower
investigators

FinTech



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Problem Statement



Problem Overview

Financial criminals exploit mule accounts to transfer and layer illicit funds across multiple bank accounts, making suspicious money flows difficult to detect and trace in time.

Why Does It Matter

- Funds can move through multiple accounts before detection.
- Complex transaction networks make manual investigations difficult.
- Delays can reduce the chances of recovering stolen funds.
- False-positive alerts waste investigators' time and can affect legitimate customer

Who Is Affected?

- Criminals use mule accounts to transfer and conceal illicit funds.
- Fraud victims may lose money through digital financial scams.
- Banks face challenges identifying suspicious transactions and connected accounts.
- AML investigators and law-enforcement agencies need to trace money trails efficiently.

Supporting Data / Statistics

- **₹55,050+ crore**: Financial-fraud amount reported through India's National Cyber Crime Reporting Portal during 2021–2025.
- **65.89 lakh+**: Financial-fraud complaints reported during 2021–2025.
- **89.2%**: Share of the value of bank frauds reported in FY 2023–24 that related to incidents from previous financial years.



The Core Challenge

- Digital payment fraud creates significant challenges for financial institutions.
- Criminals may route funds through several accounts to obscure their origin.
- Traditional rule-based alerts can miss complex relationships between accounts.
- Investigators need explainable risk scores and transaction-network visualisation to support timely decisions.

| EXISTING CHALLENGES / PROBLEM GAP



. What Is Already Being Done?

- ✓ Rule-based detection: Uses predefined rules and transaction thresholds.
- ✓ Machine learning: Predicts risk based on transaction and account features.
- ✓ Graph analytics: Analyses connections between accounts and money flows.

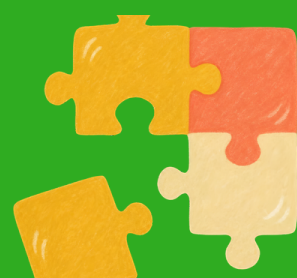
EXISTING APPROACHES

- ✓ Rule-based systems use predefined fraud-detection rules.
- ✓ ML models predict risk using transaction patterns.
- ✓ Graph analytics identify connections between accounts.



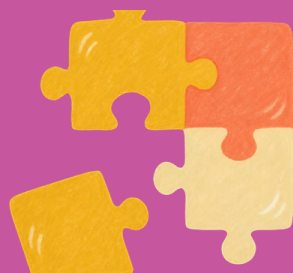
Current Pain Points and Limitations

- ✓ Rules may miss new fraud patterns.
- ✓ ML predictions may lack clear explanations.
- ✓ Manual investigations take time.



Where Do Existing Approaches Fall Short?

- ✓ Multi-hop money trails are difficult to trace.
- ✓ Separate tools complicate investigations.
- ✓ Legitimate merchants may be incorrectly flagged.



IDENTIFIED GAP

- ✓ Combine rules, ML and graph analytics in one platform.
- ✓ Trace suspicious funds across connected accounts.
- ✓ Provide explainable risk scores and reduce false positives.



Solution Overview & Core Approach

- FinGuard is an intelligent platform for detecting and investigating potential money mule networks.
- It combines rule-based detection, XGBoost risk scoring and NetworkX graph analytics.
- It provides an interactive dashboard with risk explanations and money-trail visualisation.

How the Solution Solves the Problem

- ✓ **Detect:** Identify suspicious accounts using transaction patterns and hybrid risk scores.
- ✓ **Trace:** Discover connected accounts and follow potential money flows across multiple hops.
- ✓ **Explain:** Present risk factors and supporting evidence to help investigators make informed decisions.

How FinGuard Works



| INNOVATION & UNIQUE VALUE PROPOSITION



What Is Genuinely New or Improved?

- ✓ Combines rule-based detection, XGBoost and graph analytics.
- ✓ Traces suspicious money movement across connected accounts.
- ✓ Provides explainable risk scores with evidence-based AI summaries.



What Makes It Different?

- ✓ Analyses account networks instead of isolated transactions alone.
- ✓ Combines risk prediction and money-trail investigation in one dashboard.
- ✓ Uses merchant and GST-related context to help reduce false positives.



Unique Value Proposition (USP)

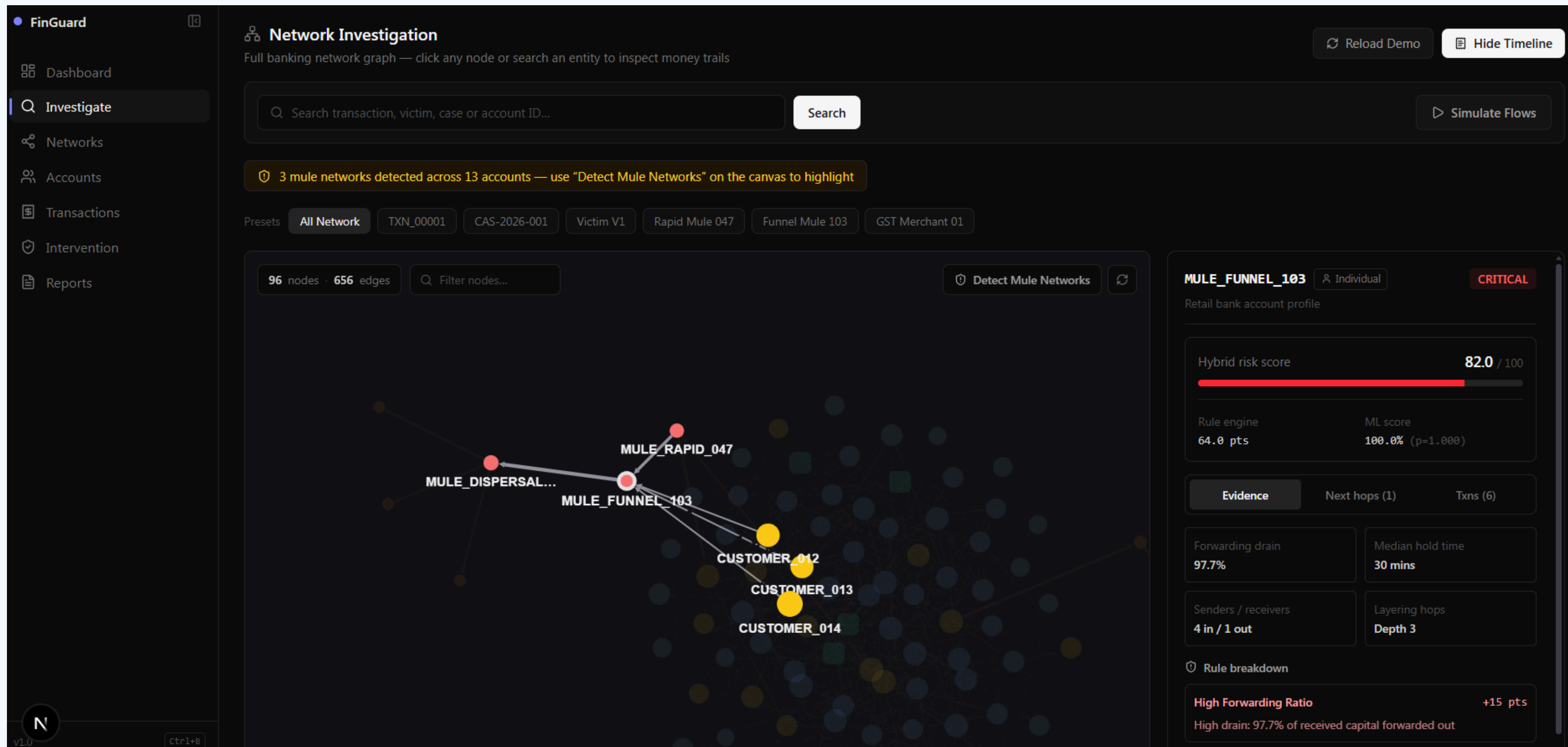
- ✓ Detect: Identify potentially suspicious accounts using hybrid risk scoring.
- ✓ Trace: Visualise connected accounts and suspicious transaction paths.
- ✓ Explain: Show risk factors and investigation evidence in one platform.



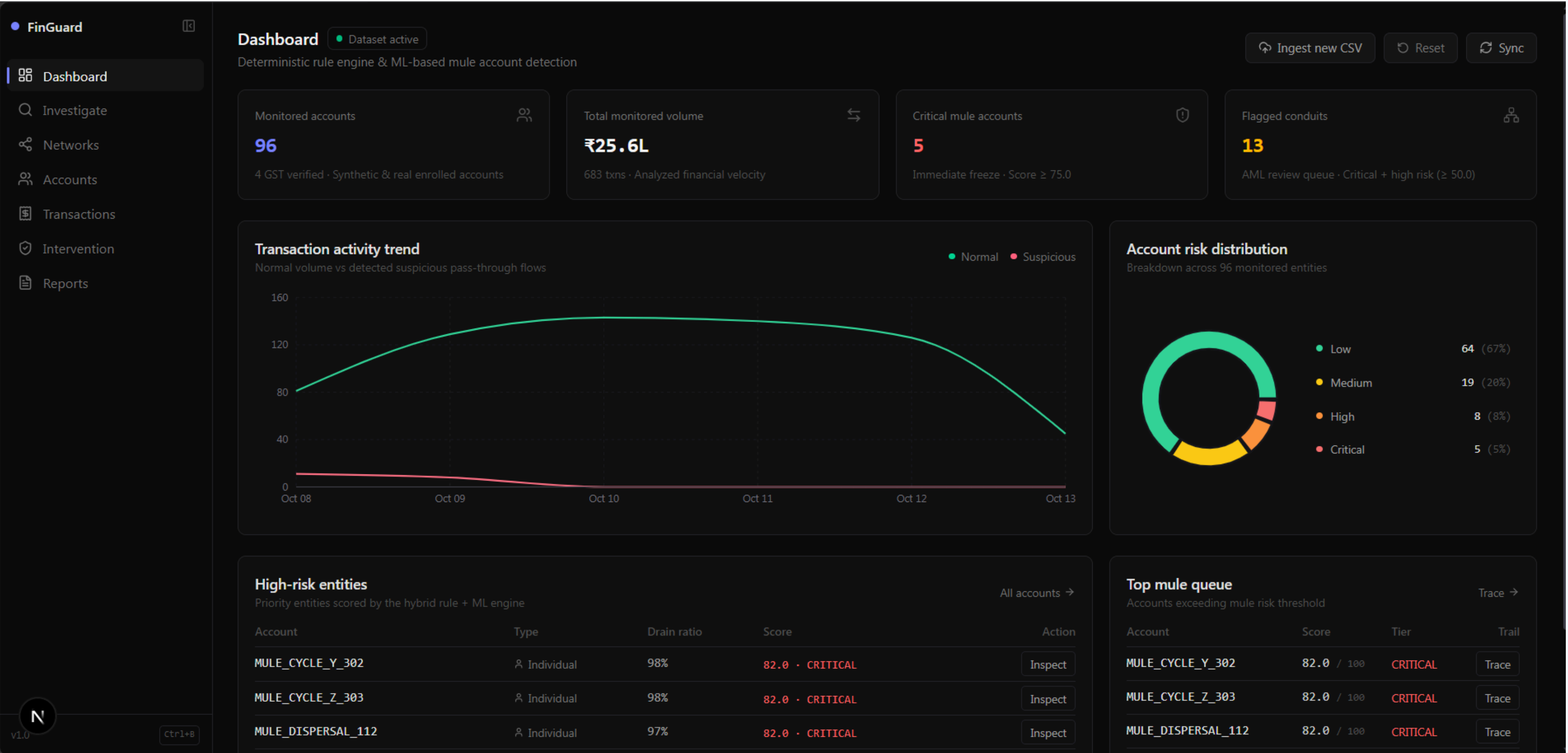
Why Should Users Choose FinGuard?

- Helps investigators prioritise high-risk accounts.
- Simplifies complex transaction-network analysis.
- Supports faster, evidence-based investigation decisions.

How the Solution Works



How the Solution Works



Tech Stack



Frontend



NEXT.JS

React
Framework



**Tailwind
CSS**

CSS Framework

Backend



Python

Programming
Language



FastAPI

Python
Framework

Database



SQLite

Relational
database

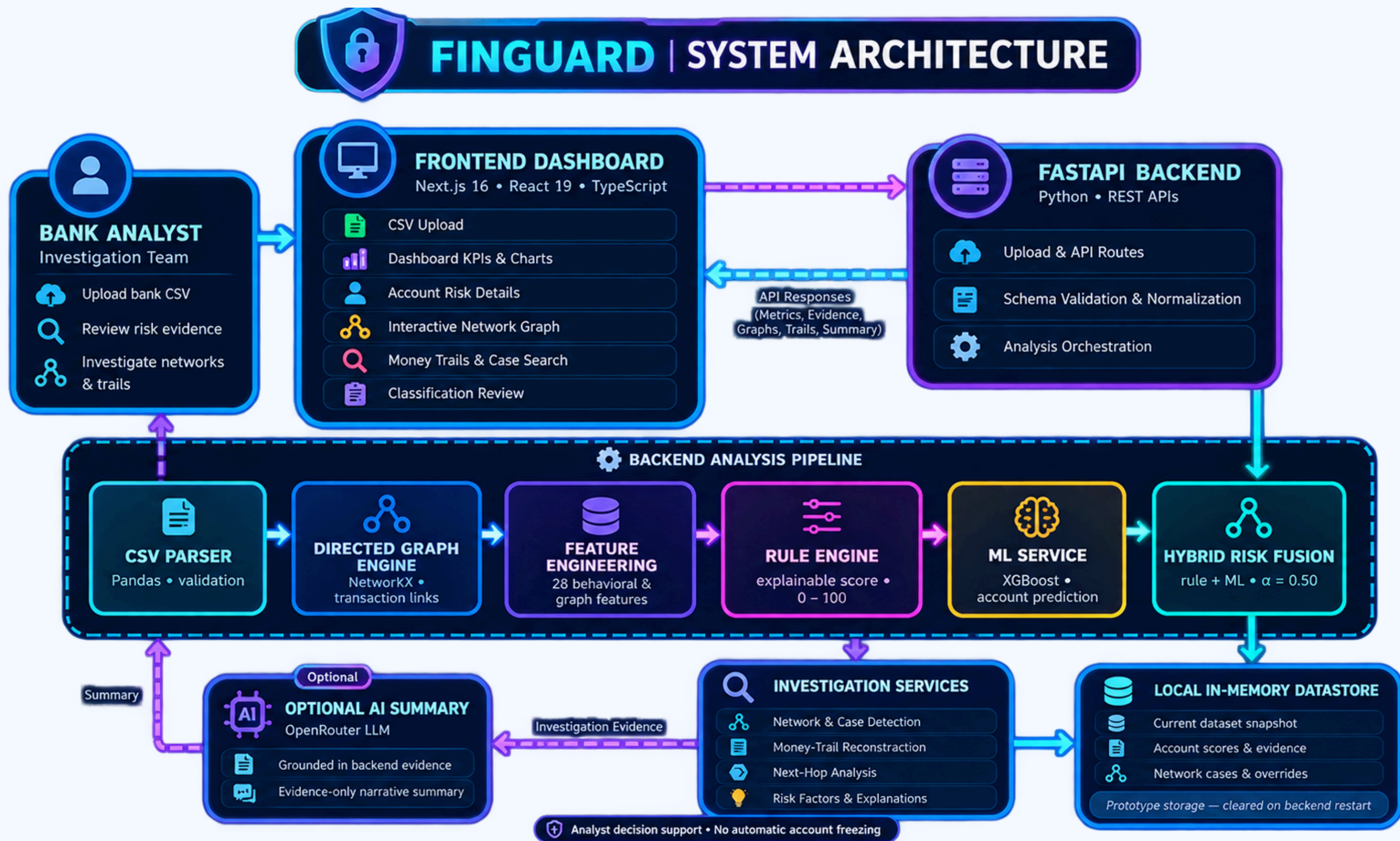
Machine Learning

XGBoost

XGBoost

Machine
Learning
library

System Architecture





| TECHNICAL FEASIBILITY & IMPLEMENTATION PLAN

Can It Actually Be Built? + Implementation Approach

- Built using Next.js, FastAPI, Python and NetworkX.
- Uses XGBoost and rule-based logic for hybrid risk scoring.
- Processes transaction CSVs and visualises suspicious networks.
- Uses modular APIs to connect the dashboard and analysis engine.

Major Technical Challenges + Mitigation

- Poor data quality: Validate CSVs and handle missing fields.
- False positives: Incorporate merchant context and explainable risk factors.
- Complex networks: Use graph-based tracing and prioritised investigations.
- AI/API failures: Keep core detection operational without optional AI summaries.

24-HOUR DEVELOPMENT STRATEGY + MILESTONES

- Hours 0–6: Set up frontend, backend and CSV ingestion.
- Hours 6–12: Implement risk scoring and graph analysis.
- Hours 12–18: Build dashboard, network visualisation and case views.
- Hours 18–24: Integrate components, test scenarios and prepare the demo.

Prototype / Demo Status

- Frontend: Next.js dashboard for investigation workflows.
- Backend: FastAPI endpoints for transaction analysis.
- Detection: Rule engine, XGBoost scoring and NetworkX graph analysis.
- Demo: Upload transaction data, inspect risk scores, visualise connections and trace money trails.

Future Scope



Future Features & Enhancements

- ✓ Real-time transaction monitoring and instant fraud alerts.
- ✓ Advanced mule-network detection and automated case prioritisation.
- ✓ Database storage for investigation history and audit trails.



Additional Integrations / AI Improvements

- ✓ Integrate with banking systems and AML platforms.
- ✓ Improve ML accuracy using validated transaction datasets.
- ✓ Generate clearer, evidence-based AI investigation summaries.



Expansion to More Languages,

- ✓ Add support for multiple Indian languages.
- ✓ Enable role-based access for analysts and compliance teams.
- ✓ Expand to more banks, regions and transaction types.



Long-Term Deployment Possibilities

- Deploy on secure cloud or bank-managed infrastructure.
- Scale to process large transaction volumes and multiple users.
- Implement stronger security, monitoring and regulatory compliance.



KEY TAKEAWAY

- ✓ FinGuard transforms complex transaction data into actionable financial intelligence.
- ✓ It helps identify suspicious accounts and trace potentially fraudulent funds.
- ✓ It combines explainable risk scoring with transaction-network analysis.



Three Strongest Benefits

- ✓ Detect: Identify potentially suspicious accounts using hybrid risk scoring.
- ✓ Trace: Visualise connected accounts and follow money trails
- ✓ Explain: Provide evidence-backed risk assessments for investigators.



LONG-TERM IMPACT

- ✓ Enable faster and more informed financial-crime investigations.
- ✓ Reduce investigative effort and help minimise false-positive alerts.
- ✓ Build a scalable, intelligent platform to strengthen financial security.